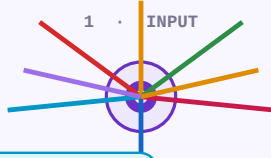


HEP Autoencoder — Particle Anomaly Detector

ATLAS Open Data · Unsupervised anomaly detection · Transformer + MLP architecture

d=128 · n_heads=4 · n_enc=4 · K=8 latent queries · latent_dim=8 · 64-dim bottleneck · Adam lr=1e-4 · 70 epochs



$\log(pT) \cdot \eta \cdot \sin\phi \cdot \cos\phi \cdot \log(m) \cdot \text{EMFrac} \cdot \text{Width} \cdot \tau_1 \cdot \tau_2$ | HT · MET_sumet · Njets · Nbjets · Nleptons ...

2 · MASKING (training only)

MASK_RATE_P = 60%

random · block-by-type · leading-3 · trailing-3 | MASK_RATE_E = 30% event features →

3 · ENCODER — d_model=128 · 4 SAB layers

Particle Embedding

MLP(phys·feat_mask) + MLP(feat_mask)
+ Embed(TYPE_VOCAB=10, 128) → LayerNorm

Event Embedding

MLP(scalar × mask) + PosEmbed(n_ev, 128)
→ LayerNorm [B, n_ev, 128]

concat → [B, n_ev + maxN, 128]

SAB #1 · MultiHeadAttention (h=4, d_head=32) + edge bias ($\Delta\eta$, $\Delta\phi$, ΔR , log m_ij) + FFN(512, GELU) + Pre-LN

SAB #2 · same structure

SAB #3

SAB #4 → output [B, n_ev + maxN, 128]

4 · BOTTLENECK — K=8 queries × latent_dim=8 = 64-dim latent

Cross-Attention Pooling · K=8 learned queries · MHA(h=4) → LayerNorm → Dense(128→8)

[B, 8] = 64-dimensional latent representation

5 · DECODER — MLP only · no cross-attention · n_dec=3

z_flat tiled × maxN + sinusoidal PE → input_proj → 3 hidden MLP blocks

feat_head

PHYS_DIM=9 outputs
softplus-tanh-sigmoid

type_head

9 class logits
8 types + no-particle

presence_head

1 logit / slot
BCE sigmoid

event_head

attn-pool over z
→ n_ev scalars

L_feat

Huber($\delta=1$) · feat_boost=[1,20,10,10,1,1,1,5,5]
active = pad_mask · feat_mask · upweight masked × (1+ α)

L_ev

Huber on event scalars · $\lambda_{EV} = 1.0$
upweight masked event slots

L_type

CrossEntropy 9 classes · $\lambda_{TYPE} = 0.5$
target = type_id or N_TYPES(no-particle)

L_pres

BCE sigmoid · $\lambda_{PRES} = 0.5$
one logit per slot · upweight masked

L_phi ϕ

ϕ -rotation equivariance · $\lambda_{PHI} = 0.05$
decode(encode($R_\alpha x$)) = R_α decode(encode(x))

L_eta η

η -flip equivariance · $\lambda_{ETA} = 0.05$
sym_w ramps 0→1 over first 30% of training

Total L = L_feat + $\lambda \cdot L_{ev}$ + $\lambda \cdot L_{type}$ + $\lambda \cdot L_{pres}$ + sym_w · ($\lambda \phi \cdot L_\phi$ + $\lambda \eta \cdot L_\eta$)

sym_w ramps from 0 → 1 over warmup steps · gradient clipping: global_norm ≤ 5.0